UNIVERSITÀ DEGLI STUDI FIRENZE

Scuola di Scienze Matematiche, Fisiche e Naturali

Laurea Magistrale in Data Science, Calcolo Scientifico & Intelligenza Artificiale

Master Thesis

THESIS' TITLE

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Academic Year 2023-2024

ABSTRACT

Nowadays the huge amount of data generated by digital financial systems, together with the evolution of fraud schemes, make fraud detection a challenging and critical problem. Since illicit actors employ sophisticated strategies such as fraud rings and cyclic transfers to obscure their trails, traditional heuristic and rule-based detection systems frequently fail to provide an effective solution.

The primary goal of this thesis is to develop a detailed Python implementation of a Quantum Topological Graph Neural Network (QTGNN) [3]. This model combines graph neural networks, quantum state embedding and topological data analysis to capture higher-order, long-range dependencies and robust topological invariants that are typical in fraud schemes. Leveraging these features we will also be able to enhance the interpretability of the model. This is extremely important since, in regard to using deep learning models in the real world and particularly in legal settings, the model must be able to explain its decisions [4].

The model will be evaluated on four different transaction datasets: the Elliptic Bitcoin Dataset, Ethereum Phishing Transaction Networks, IBM AMLSim, and PaySim. To balance computational feasibility with practical applicability, the model will be trained on a quantum simulator. Only the final testing phase will be executed on an actual NISQ quantum computer.

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INTRODUCTION

1.1 CONTEXT AND IMPORTANCE

Fraud detection has become a foundational pillar of modern economic security. The rapid digitization of financial services has streamlined global commerce but has also opened new vectors for illicit activities. Undetected financial fraud leads to massive economic losses for individuals and institutions, undermines public trust in the financial system, and frequently serves to finance broader criminal organizations, including money laundering and terrorism. Consequently, developing robust mechanisms to identify and halt fraudulent transactions is critically important to preserving both market integrity and social stability.

1.2 THE GROWING CHALLENGE

Detecting fraud is becoming increasingly difficult. The sheer volume and velocity of digital transactions make manual oversight impossible and heavily strain classical machine learning architectures. Furthermore, the rise of decentralized finance (DeFi), Web3, and cryptocurrencies has introduced a paradigm of pseudo-anonymity that complicates traditional tracing.

Criminals have adapted to this landscape by employing highly sophisticated obfuscation tactics. Instead of acting alone, they form complex "fraud rings" to distribute illicit funds across vast networks of temporary accounts. To further break the traceable link between sender and receiver, illicit actors frequently use cyclic transfers and cryptocurrency **mixers**—services that pool together funds from multiple users, mix them, and redistribute them at random intervals and amounts. These advanced tactics create dense, highly camouflaged transaction topologies that easily evade threshold-based rules and simple heuristic detection methods.

1.3 PROPOSED SOLUTION: QTGNN

Traditional Graph Neural Networks (GNNs) have shown promise in fraud detection by utilizing local message-passing mechanisms. However, they often struggle to capture the long-range dependencies and higher-order topological structures that characterize modern, sophisticated fraud rings.

To address these limitations, this thesis proposes the use of a Quantum Topological Graph Neural Network (QTGNN). By combining the principles of Quantum Computing, Graph Theory, and Topological Data Analysis (TDA), the QTGNN offers a remarkably robust solution. It leverages quantum embeddings to capture multiple complex transaction pathways simultaneously and employs topological invariants (such as persistent homology) to identify the underlying structural "shape" of the fraud network, isolating it from deliberate noise and obfuscation.

1.4 THESIS OBJECTIVES

The primary objective of this thesis is to develop a comprehensive Python implementation of the QTGNN specifically tailored for fraud detection.

To evaluate the model's performance against varying structural challenges, the implementation will be tested on several distinct datasets:

- **The Elliptic Bitcoin Dataset**: A real-world dataset mapping Bitcoin transaction flows that contains rich topological structures obfuscated by illicit actors.
- **Ethereum Phishing Transaction Networks**: A real-world dataset where accounts and transactions are modeled as nodes and edges to perform phishing account classification.
- **IBM AMLSim**: A dataset containing synthetic everyday transactions embedded with specific alert patterns and anomalies that mimic actual money laundering typologies, such as smurfing.
- **PaySim**: A synthetic financial dataset for evaluating broader classification capabilities.

Given the current limitations of quantum hardware, this research adopts a hybrid testing strategy. The QTGNN model will be trained heavily using a quantum simulator, while the final evaluation and testing will be performed on an actual quantum computer to effectively reduce computational costs and validate its viability on Noisy Intermediate-Scale Quantum (NISQ) devices.

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FRAUD RINGS THEORY

2.1 THE LIMITATIONS OF THE LONELY SCAMMER

Individual scammers typically rely on simple and intuitive techniques to introduce illicit funds into the legitimate economy. One of the most prevalent methods is known as "smurfing". In simple terms, smurfing involves breaking down a large sum of illicit money into many smaller, inconspicuous chunks to deposit them without raising suspicion.

More formally, individual scammers rely on smurfing to avoid Anti-Money Laundering (AML) detection mechanisms, which usually flag transactions above a specific threshold $\tilde{x}$. By splitting their initial illicit funds x into smaller transactions x_n, such that each transaction satisfies $x_n \leqslant \tilde{x}$, they minimize the risk of detection [2]. The mathematical expectation of profit π_s for a scammer engaging in smurfing can be defined as:

$$\pi_s = x(1 - \nu) \tag{2.1}$$

where x represents the total initial illicit funds, and $\nu \in (0, 1)$ is the friction cost or commission proportion associated with the smurfing process (e.g., the fee paid to mules).

While smurfing is effective for small amounts, its scalability is severely limited. When attempting to launder much larger volumes of funds, an individual scammer operating from a single account faces a significantly higher probability of detection $(1 - p)$, where p is the probability of successfully avoiding detection. If caught, the scammer is subjected to a severe monetary penalty T, which includes both the total confiscation of the funds and associated judicial fines [2]. Therefore, operating as a "lonely scammer" becomes mathematically and practically unviable for large-scale operations due to the massive expected cost of $(1 - p)T$.

2.2 THE EMERGENCE OF FRAUD RINGS

To circumvent the limitations of individual operations and avoid getting caught, scammers evolved to form collusive networks, commonly referred to as "fraud rings". These rings operate by dispersing the illicit funds across a vast, interconnected network of accounts. They utilize a strategy of transaction layering, often characterized by a topological depth L, and network diversification to obscure the money trail [2, 5].

By spreading the funds through a complex topology, the fraud ring successfully increases its non-detection probability p_L, which now depends explicitly on the layering depth L. The expected profit of a fraud ring utilizing L layers can be mathematically expressed as:

$$E[\pi_L] = p_L x - (1 - p_L)T - c(L) \tag{2.2}$$

where $E[\pi_L]$ is the expected profit, p_L is the enhanced probability of evading detection when L layers are used, x is the total illicit funds, T is the total penalty if caught, and $c(L)$ represents the monotonic cost function associated with establishing and maintaining the L transaction layers. Because the fraud ring dramatically increases the probability of evading detection ($p_L \to 1$ as L increases), the expected payoff remains highly profitable even when factoring in the complex layering costs $c(L)$ [2].

2.3 GRAPH THEORY, GNNS, AND QUANTUM COMPUTING

Because fraud rings create such complex, multi-layered transaction topologies, fraudulent behaviors can propagate with remarkable speed and sophistication [1]. Traditional heuristic or rule-based detection tools—which often rely on simple random sampling or threshold rules—fail to identify them effectively. To counter these sophisticated rings, modern detection systems leverage Graph Theory and deep learning [5].

The transaction network is modeled as a directed graph $G = (V, E)$, where V represents the set of accounts (nodes) and E represents the set of transactions (edges). While classical Graph Neural Networks (GNNs) have proven highly successful in analyzing local neighborhood dependencies through message-passing mechanisms, they often fail to capture the higher-order, long-range dependencies critical for identifying distributed and coordinated fraudulent behaviors [7]. One technique to sample and study these dense, evolving networks is through temporal random walk subgraphs, which allow models to capture the dynamic sequence of transactions over time (this sampling mechanism will be detailed thoroughly in a subsequent implementation section).

In this work, we will utilize a Quantum Topological Graph Neural Network (QTGNN) to learn the dense, anomalous connectivity inherent to fraud rings. Drawing from recent advancements [3], QTGNN bridges quantum machine learning, graph theory, and topological data analysis, working exceptionally well for this task by addressing three fundamental challenges:

- **Leveraging quantum embeddings for richer relational representation:** Traditional representations struggle to model the high-dimensional pathways of fraud rings. Quantum embeddings encode this complex relational information by exploiting quantum superposition and entanglement. This allows the model to represent multiple transaction pathways and interdependencies simultaneously, capturing subtle patterns distributed across the network that classical local message-passing GNNs often overlook [3].

- **Incorporating topological invariants for robustness and interpretability:** Topological Data Analysis (TDA) has gained significant attention for its ability to extract invariant features that characterize the global structure of data [6]. By extracting topological descriptors such as persistent homology, the QTGNN captures global structural features that remain invariant to noise and minor perturbations. This not only makes the representations robust against the spurious connections generated by scammers but also enhances the model's interpretability by identifying the higher-order motifs driving the anomaly detection.

- **Ensuring scalability and practicality on NISQ devices:** While quantum models offer theoretical advantages, their practical deployment is heavily constrained by the limitations of Noisy Intermediate-Scale Quantum (NISQ) hardware [9]. The QTGNN framework addresses these bottlenecks through a NISQ-aware design that employs circuit optimization, error mitigation, and hybrid quantum-classical computation strategies. This ensures that the quantum-enhanced graph learning remains feasible, scalable, and fully deployable on real-world financial transaction networks [3].

A comprehensive mathematical explanation of the QTGNN architecture and its implementation will be provided in a future section.

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